Exploratory Data Analysis (EDA) Summary Report Template

1. Introduction

The purpose of this report is to conduct an exploratory data analysis (EDA) on Geldium's customer dataset to assess data quality, identify missing or inconsistent values, and detect early risk indicators related to credit card delinquency.

The analysis aims to support Tata iQ's analytics team in preparing reliable data for future predictive modeling and AI-driven risk assessment.

2. Dataset Overview

The dataset contains customer financial, behavioral, and demographic information used to assess delinquency risk. The analysis identified several important variables associated with repayment behavior and potential financial stress.

<u>Key dataset attributes:</u>

- Number of records: Approximately 290+ customer records

- Key variables:

  • Income – Annual customer income

  • Credit_Score – Customer credit worthiness score

  • Credit_Utilization – Percentage of available credit used

  • Missed_Payments – Number of missed payments

  • Debt_to_Income_Ratio – Debt obligations compared to income

  • Loan_Balance – Outstanding loan amount

  • Employment_Status – Employment category

  • Delinquent_Account – Indicates delinquency status

  • Month_1 to Month_6 – Monthly payment history

- Data types:

  • Numerical: Income, Credit_Score, Credit_Utilization, Debt_to_Income_Ratio, Loan_Balance

  • Categorical: Employment_Status, Credit_Card_Type, Location, Monthly payment statu

3. Missing Data Analysis

Several missing values and inconsistencies were identified during the analysis. Missing data was primarily observed in Income, Loan_Balance, and Credit_Score fields, which are important variables for delinquency prediction.

<u>Key missing data findings:</u>

- Variables with missing values:

• Income

• Loan_Balance

• Credit_Score

- Missing data treatment:

• Income values were planned for median imputation to reduce the influence of extreme income values.

• Missing Loan_Balance values may be handled using synthetic data generation or statistical imputation to maintain dataset completeness.

• Missing Credit_Score values may be imputed using mean or median values based on overall dataset distribution.

- Additional data quality issues:

• Inconsistent Employment_Status formatting such as “EMP”, “Employed”, and “employed”

• Extremely high Credit_Utilization values above 1.0 indicating possible outliers or abnormal borrowing behavior

4. Key Findings and Risk Indicators

The analysis identified several financial and behavioral patterns associated with delinquency risk.

<u>Key findings:</u>

- Correlations observed between key variables:

• Customers with higher Missed_Payments frequently showed delinquent account behavior.

• High Credit_Utilization was associated with increased repayment risk and financial stress.

• High Debt_to_Income_Ratio indicated possible difficulty managing financial obligations.

• Repeated “Late” and “Missed” payment patterns across Month_1 to Month_6 were strong indicators of delinquency risk.

- Unexpected anomalies:

• Some customers with high Credit_Score values were still marked as delinquent, suggesting additional behavioral factors influence repayment risk.

• Certain customers showed unusually high Credit_Utilization values greater than 1.0.

• Inconsistent categorical labels may affect predictive model performance if not standardized.

5. AI & GenAI Usage

Generative AI tools were used to summarize the dataset, impute missing data, and detect patterns. This section documents AI-generated insights and the prompts used to obtain results.

Example AI prompts used:

-“Summarize key patterns, outliers, and missing values in this dataset.”

- "Identify the top 3 variables most likely to predict delinquency.”
- "Suggest an imputation strategy for missing income values based on industry best practices.”

- "Analyze trends in payment history and identify major delinquency risk indicators."

6. Conclusion & Next Steps

The exploratory data analysis identified several important risk indicators related to delinquency, including missed payments, high credit utilization, and high debt-to-income ratios.

Missing values and inconsistent formatting issues were also detected in multiple fields, which may affect predictive modeling accuracy if not properly handled.

The dataset provides strong foundational information for future AI-driven delinquency prediction models after appropriate preprocessing and data cleaning.

The next step will involve preparing the cleaned dataset for predictive modeling and evaluating customer risk patterns in greater detail.